

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
NASA SHARED SERVICES CENTER

RECOMMENDATION AND DETERMINATION TO SOLICIT FROM ONE SOURCE
OR BRAND NAME

PICK ONE

Sole Source:

☒ I recommend that NASA, NASA Shared Services Center negotiate with Texas A&M Engineering Experiment Station only for Shape memory alloy radiator development, modeling, and research in support of PaCeSS IR&D. The total estimated cost of this effort is [REDACTED] and the estimated period of performance or lead-time for delivery is No later than 9/30/2025.

OR

Brand Name Justification:

☐ I recommend that NASA, NASA Shared Services Center negotiate with only with vendors who can provide the subject brand name from Manufacture Name for Enter Supply or Service to be procured. The total estimated cost of this effort is Enter Dollar amount and the estimated period of performance or lead-time for delivery is Date.

This recommendation is for the acquisition of supplies or services determined to be reasonably available from only one source/only one manufacture. Competition is impractical for the following reasons (please answer the questions below):

1. **What are the reasons for soliciting only the suggested source/suggested brand name?** Please provide the unique characteristics of the items or services along with the supporting rationale and the market research performed to substantiate your reason for the suggested source/brand name.

The requested procurement develops a technology that is unique to and patented (US10228197B2) in part by Texas A&M University (TAMU). The technology is unique and critical to the Passive fuel-Cell Surface-power System (PaCeSS) effort as an element that controls the temperature of the fuel cell without active control or wearing parts. This procurement leverages prior work by Texas A&M that a) demonstrates actuation of a Shape memory torque tube actuator with a two-phase working fluid, b) models and predicts the behavior of shape memory materials, and c) torque tube materials on hand that operate in the temperature range of interest.

2. **What is the impact to the government if another vendor/brand is used?** Include any dollar, schedule, or compatibility impacts if possible.

Estimated Impacts if another vendor used: Schedule – impacts period of performance by at least 12 months; [REDACTED] Schedule impact justification: equivalent to SBIR phase I (1yr), R&D effort needed to develop innovation comparable to requested sole source and procurement time for torque tube materials that TEES has on hand. [REDACTED]

ADDITIONAL COMMENTS:
